

DIFFUSION OF NICKEL INTO IRON*

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The diffusion of nickel into iron was measured in the temperature range 600–1050° C. Radioactive Ni^{63} was used as the tracer element; the surface-decrease and the residual-activity sectioning methods were employed. The diffusivity below the Curie temperature was observed to be lower than that expected from an extrapolation of the diffusion data for paramagnetic alpha-iron.

The diffusion coefficients may be expressed as follows (in cm^2/sec):

$$\text{Gamma-iron, } D = 0.77 \exp(-67,000/RT)$$

$$\text{Paramagnetic alpha-iron above } 800^\circ \text{C, } D = 1.3 \exp(-58,000/RT)$$

$$\text{Ferromagnetic alpha-iron below } 680^\circ \text{C, } D = 1.4 \exp(-58,700/RT).$$

The anomalous decrease in the diffusion coefficient starts at about 800° C, somewhat above the Curie temperature, and is thought to be associated with the effect of short range magnetic order on the formation energy of vacancies.

DIFFUSION DU NICKEL DANS LE FER

La diffusion du nickel dans le fer a été mesurée pour la gamme de températures s'étendant de 600 à 1050°C. Le Ni^{63} radio-actif est utilisé comme traceur; les méthodes employées sont celles de la diminution de la surface et de l'activité résiduelle. La diffusibilité en-dessous de la température de Curie est inférieure à ce que l'on pourrait attendre de l'extrapolation des données expérimentales de la diffusion dans le fer α paramagnétique. Les coefficients de diffusion peuvent s'exprimer comme suit (en cm^2/sec).

$$\text{Fer gamma, } D = 0,77 \exp(-67,000/RT)$$

$$\text{Fer alpha paramagnétique au-dessus de } 800^\circ \text{C, } D = 1,3 \exp(-58,000/RT)$$

$$\text{Fer alpha ferromagnétique en-dessous de } 680^\circ \text{C, } D = 1,4 \exp(-58,700/RT)$$

La diminution anormale du coefficient de diffusion débute aux environs de 800°C, c'est-à-dire un peu au-dessus de la température de Curie. Elle peut être associée à l'effet de l'ordre magnétique à petite distance sur l'énergie de formation des lacunes.

DIFFUSION VON NICKEL IN EISEN

Die Diffusion von Nickel in Eisen wurde im Temperaturgebiet 600–1050°C gemessen. Als Spurenelement wurde radioaktives Ni^{63} benutzt; dabei wurde sowohl die Abnahme der Aktivität an der Oberfläche wie die Restaktivität von Schnitten der Probe zur Messung verwandt. Die Diffusionskonstante unterhalb der Curietemperatur ist niedriger, als man es nach Extrapolation der Diffusionsmessungen für paramagnetisches Alpha-Eisen erwartet.

Die Diffusionskoeffizienten lassen sich folgendermaßen ausdrücken (in cm^2/sec):

$$\text{Gamma-Eisen } D = 0.77 \exp(-67,000/RT)$$

$$\text{Paramagnetisches Alpha-Eisen über } 800^\circ \text{C } D = 1.3 \exp(-58,000/RT)$$

$$\text{Ferromagnetisches Alpha-Eisen unter } 680^\circ \text{C } D = 1.4 \exp(-58,700/RT)$$

Die anomale Abnahme des Diffusionskoeffizienten beginnt bei etwa 800°C, etwas oberhalb der Curietemperatur, und hängt vermutlich mit dem Einfluss der magnetischen Nahordnung auf die Bildungsenergie von Leerstellen zusammen.

1. INTRODUCTION

The diffusivity of nickel in gamma-iron has been measured by Wells and Mehl⁽¹⁾, but apparently the diffusion of nickel in alpha-iron has not been investigated. Recent work has shown that the self-diffusivity of alpha-iron below the Curie temperature is markedly lower than that expected from diffusion measurements on paramagnetic alpha-iron.^(2,3) The present investigation was undertaken to provide diffusion data for nickel in iron, with numerous determinations being made in the vicinity of the Curie temperature to

obtain an independent check on the influence of a magnetic transformation on the diffusivity.

2. EXPERIMENTAL PROCEDURE

High-purity iron‡ was obtained in the form of $\frac{1}{8}$ -in. dia. forged bars, which were then machined to $\frac{3}{8}$ -in. dia. The bars were sealed in evacuated Vycor tubes and annealed at 850°C for one week. The resultant grain size was $\frac{1}{10}$ – $\frac{1}{4}$ in. A few single crystals were also made by a gradient anneal of bars that had been tensile strained through the discontinuous yielding. Disc specimens of $\frac{1}{2}$ -in. dia. and $\frac{3}{8}$ -in. thickness were machined from the bars and the

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‡ The analysis showed (in wt. %): 99.97 Fe, 0.007 C, 0.0056 O, 0.00013 N and less than 0.005 Si.

ends were ground on emery paper. The specimens were again annealed in vacuum at 880°C for 3 hr.

One mc of Ni^{63} , which decays with the emission of 0.063 MeV β particles and has a half life of 93 years, was received from Oak Ridge National Laboratory in the form of an acid chloride solution. Thin layers of the isotope, about 0.1μ thick, were electroplated onto the flat faces of the specimens from a solution of nickel sulfate, ammonium chloride, and boric acid containing both radioactive and inert nickel ions. The specimens were sealed in evacuated tubes and diffusion-annealed in furnaces controlled within $\pm 1^\circ\text{C}$.

The diffusion coefficients were determined by two methods. In the residual-activity technique, activity measurements were made on the disc specimen after the removal of successive layers from the surface. This method has been used with Ni^{63} by Hoffman *et al.*⁽⁴⁾ and Mackliet⁽⁵⁾. A windowless flowcounter with 2π geometry was employed in the current experiments, and more than 30,000 counts were recorded for each specimen. If I_n is the activity of the sample after a total thickness y_n has been removed, and y is the distance into the specimen as measured from the initial surface, then for this geometry:

$$I_n = C \int_0^{\pi/2} \int_{y_n}^{\infty} (\sin \theta) \times \exp \left(-\frac{y^2}{4Dt} - \mu(y - y_n) \sec \theta \right) dy d\theta \quad (1)$$

and if the absorption, μ , is high:

$$\mu I_n = K \exp(-y_n^2/4Dt) \quad (2)$$

where C and K are constants and t is the diffusion time. Thus, a plot of $\ln I_n$ vs. y_n^2 will have the same slope as the usual penetration curve; some examples are shown in Fig. 1. The validity of the residual-activity technique has been discussed by Mackliet⁽⁵⁾ who used this method for the diffusivity of Ni^{63} in copper.

For small penetrations, it was necessary to use the surface-decrease method.^(4,6-8) The final activity at the surface, I , relative to the initial value, I_0 , for a straight-line counting geometry is given as

$$I/I_0 = (1 - \text{erf}[\mu(Dt)^{1/2}]) \exp(\mu^2 Dt) \quad (3)$$

where $\text{erf } z = 2(\pi)^{-1/2} \int_0^z \exp(-z^2) dz$.

For the 2π geometry used here,

$$\frac{I}{I_0} = \frac{\int_0^{\pi/2} \exp(\mu^2 Dt \sec^2 \theta) [1 - \text{erf}(\mu(Dt)^{1/2} \sec \theta)] \sin \theta d\theta}{\int_0^{\pi/2} \frac{1 - \exp(-\mu a \sec \theta)}{\mu a \sec \theta} \sin \theta d\theta} \quad (4)$$

where a is the thickness of the thin layer from which the diffusion takes place. Because of the high absorption coefficient in the case of Ni^{63} , and since the value of μ is not known with sufficient certainty, the relationship between I/I_0 and (Dt) was determined empirically for the 2π geometry using specimens diffused at 750 and 840°C where the diffusivity had been measured by the residual-activity technique. Typical surface-decrease curves are shown in Fig. 2 as a function of (Dt) . Curve A was calculated by a graphical integration of equation (4) using a value of $\mu = 1 \times 10^4 \text{ cm}^{-1}$ ⁽⁴⁾ and assuming $a = 0$. Curve C was calculated using equation (3) and indicates the error introduced in neglecting the 2π geometry. Curve B is the empirical curve obtained by plotting measured I/I_0 ratios of 750 and 840° specimens against the $\sqrt{(Dt)}$ values, D being known from independent residual-activity experiments on the same specimens. It is evident that the empirical curves were satisfactory for the determination of diffusivities by the surface-decrease method, making it unnecessary to solve equation (4) for each run.

3. RESULTS and DISCUSSION

The diffusivities are summarized in Table I and the temperature dependence is shown in Fig. 3. Two

TABLE I. Experimental data for diffusion of nickel into iron

Temperature (°C)	Duration of diffusion anneal (sec)	D (cm^2/sec)	Technique and specimen
1050	2.16×10^5	6.01×10^{-12}	1 A
1000	2.88×10^4	2.37×10^{-12}	1 A
960	8.64×10^4	1.31×10^{-12}	1 A
930	1.73×10^5	4.25×10^{-12}	1 A
900	2.59×10^5	5.23×10^{-11}	1 B
900	2.59×10^5	4.15×10^{-11}	1 A
870	8.67×10^5	2.70×10^{-11}	1 B
870	8.67×10^5	2.73×10^{-11}	1 A
840	1.30×10^6	1.31×10^{-11}	1 B
840	1.30×10^6	1.24×10^{-11}	1 A
810	1.30×10^6	8.56×10^{-12}	1 B
810	1.30×10^6	5.26×10^{-12}	1 A
780	7.78×10^6	1.41×10^{-12}	1 A
780	7.78×10^6	1.07×10^{-12}	1 A
780	7.78×10^6	1.92×10^{-12}	1 A
780	7.78×10^6	1.83×10^{-12}	1 A
750	7.78×10^6	4.13×10^{-12}	1 A
750	7.78×10^6	3.72×10^{-12}	1 A
750	7.78×10^6	7.41×10^{-12}	1 A
750	7.78×10^6	5.55×10^{-12}	1 A
710	1.81×10^6	2.4×10^{-12}	2 A
710	1.81×10^6	1.5×10^{-12}	2 A
680	1.81×10^6	6.3×10^{-14}	2 A
680	1.81×10^6	3.5×10^{-14}	2 A
600	5.18×10^6	3.6×10^{-12}	2 A
600	5.18×10^6	2.1×10^{-12}	2 A

Techniques 1: residual-activity technique.

2: surface-decrease technique based on the empirical surface-decrease curve.

Specimens A: single crystal.

B: polycrystal of large grain size.

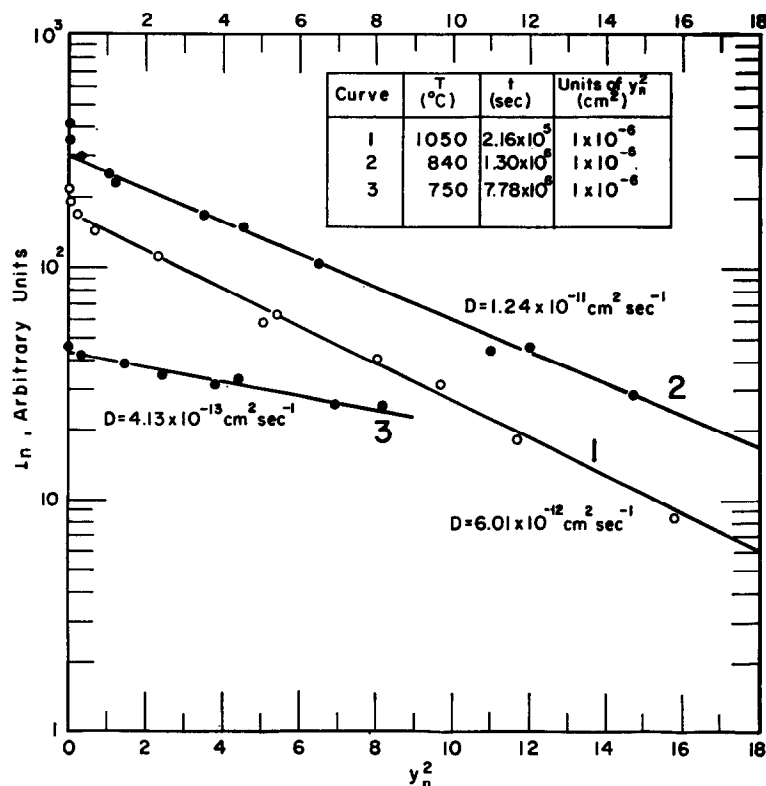


FIG. 1. Typical penetration curves for diffusion of Ni^{58} in iron.

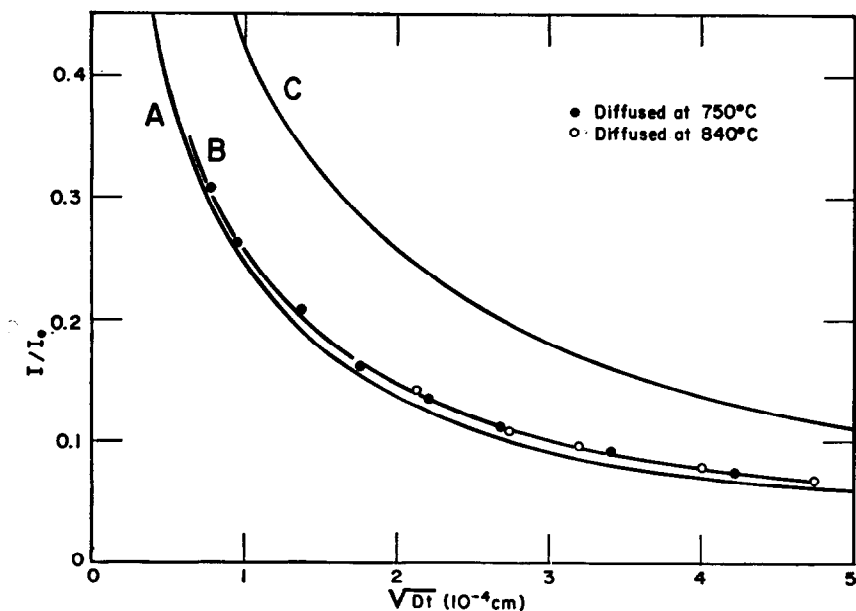


FIG. 2. Decrease in surface activity as a function of $\sqrt{Dt}$.
 Curve A: Calculated using 2π geometry (equation 4).
 Curve B: Empirical surface-decrease curve for specimens diffused at 750 and 840°C.
 Curve C: Calculated using straight-line geometry (equation 3).

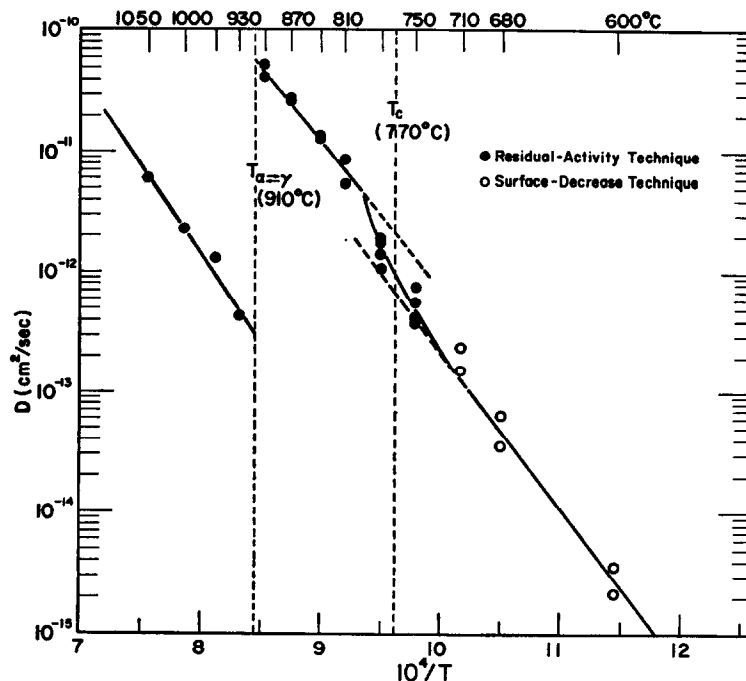


FIG. 3. Temperature dependence of diffusion coefficient of nickel into iron.

points were obtained, from opposite faces of each specimen, for the runs below 780°C. It is evident from Fig. 3 that there is a change in the diffusivity associated with the magnetic transformation in iron, confirming the previous data on self-diffusion.^(2,3) The diffusivity begins to decrease at a temperature just above the Curie temperature (770°C); apparently, this anomaly is related to the high degree of magnetic short range order in iron just above the transition.⁽⁹⁾ The anomaly cannot be ascribed to a deficiency in the surface-decrease method at the low temperatures since the residual-activity technique was used as low as 780 and 750°C where the Curie point effect is quite noticeable.

The activation energies and frequency factors were calculated by the least-squares method for three temperature ranges, and these values are listed in Table 2. An activation energy was not fitted in the anomalous range. The effect of the magnetic transition cannot be explained in terms of random or systematic errors. The total probable error in the activation energy is one per cent for the gamma-iron and less than two per cent for the paramagnetic alpha-iron. The error in D_0 is about ± 25 per cent in both cases. The error in the ferromagnetic iron range was much greater, primarily due to the small penetrations at the low temperatures.

These results are rather similar to the values obtained in this laboratory⁽³⁾ for the self-diffusion of

iron. A direct comparison of the diffusion coefficients is given in Fig. 4. In both cases, the magnetic anomaly sets in at 20–30°C above the Curie temperature. According to Table 2, and Table 3 of Ref. 2,

TABLE 2. Temperature dependence of the diffusion coefficient of nickel into iron

Experimental temperature range (°C)	D_0 (cm ² /sec)	Q (kcal/mole)
930–1050 gamma-iron	0.77	67.0
810–900 paramagnetic alpha-iron	1.3	56.0
600–680 ferromagnetic alpha-iron	1.4	58.7

the effect is associated mainly with a small increase in activation energy in the ferromagnetic region, with the frequency factor remaining about the same. Thus, there appears to be a change in the vacancy concentration or the vacancy mobility.

Considering D_0 to be constant, the ratio of the diffusion coefficient in the ferromagnetic region, D_f , to the extrapolated value from the paramagnetic region, D_p , may be written:

$$\frac{D_f}{D_p} = \exp [-(Q_f - Q_p)/RT]. \quad (5)$$

Let us assume that the difference in activation

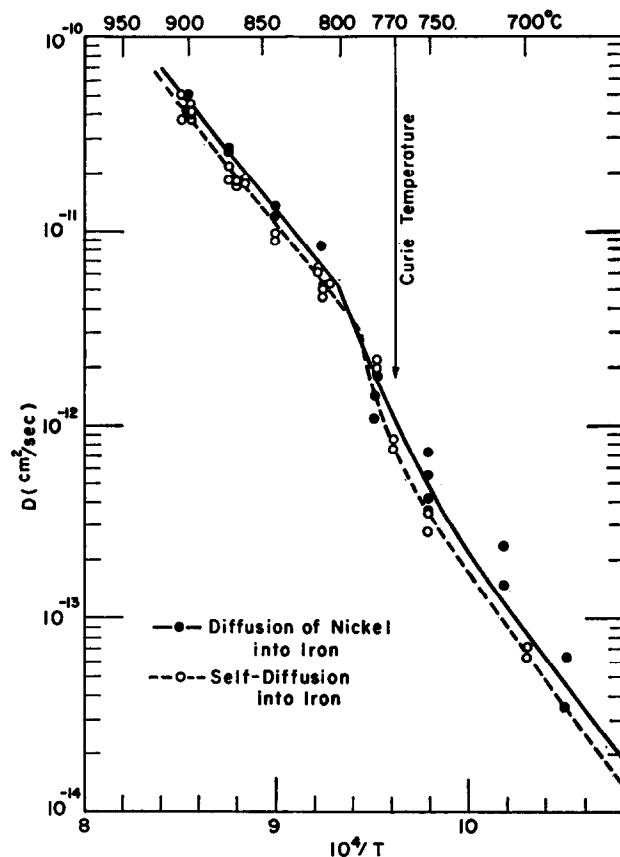


FIG. 4. Comparison of diffusivity of nickel with that for self-diffusion in iron.

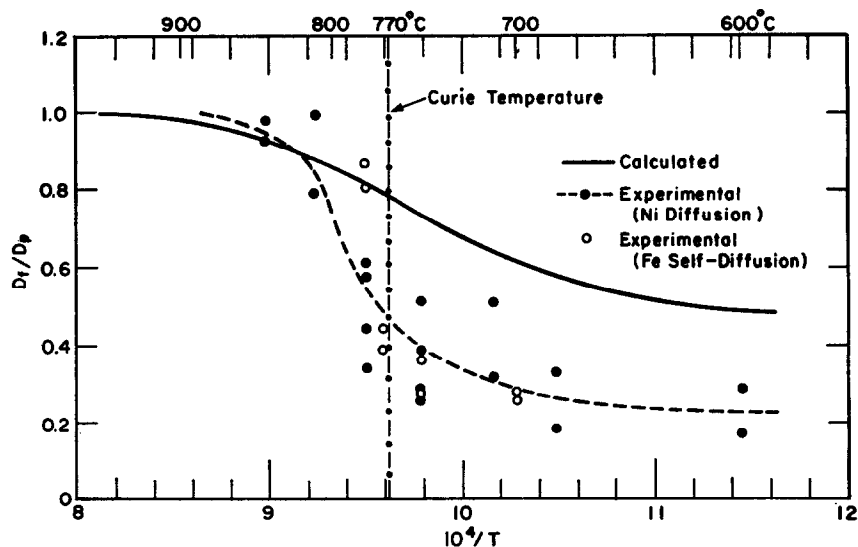


FIG. 5. The ratio of the diffusion coefficient in ferromagnetic iron to the extrapolated value for paramagnetic iron.

energies arises only from a greater formation energy of vacancies in the ferromagnetic phase, neglecting any influence on vacancy motion. Presumably, a vacancy could result in local demagnetization, which would require additional energy. The energy of magnetiza-

tion for ferromagnetic iron as a function of temperature was estimated by a graphical integration of the magnetic specific heat vs. temperature curve given by Hofmann *et al.*⁽¹⁰⁾; setting these energy values equal to $Q_f - Q_p$ in equation (5), the calculated

D_f/D_p ratios are plotted in Fig. 5. It is evident that the calculated effect is of the right order of magnitude relative to the measured D_f/D_p values. The energy of magnetization begins to play an important role on cooling just above Curie temperature and reaches a maximum of about 1.9 kcal/mol,⁽¹⁰⁾ which is comparable to the experimental value of $Q_f - Q_p = 2.7$ kcal obtained in the diffusion experiments (Table 2).

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